

Vertex crossings and boundary regimes in a symmetric Markov multinomial model

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Abstract

We study occupation counts of a stationary Markov chain that repeats its current state with probability p and otherwise chooses uniformly among $d - 1$ alternatives. For fixed d , we compare the probability of an individual count vector with that of a vertex, where every visit is to one state. A positive refresh expansion gives a relative Bessel approximation uniform over all positive compositions, including vectors approaching simplex boundaries. We obtain explicit crossing expansions for balanced faces and interior compositions, and a rare-coordinate crossover at $N/(\log N)^2$ when at least two coordinates are macroscopic. Near a vertex, a product of finite Laguerre polynomials determines the leading crossing uniformly for arbitrary rare counts with total $A = o(b)$, where b is the dominant count; the relative root error is $O_\ell(\sqrt{(A+1)/b})$. This root approximation can hold even when its corresponding probability approximation diverges. We also determine spatial superlevel volumes and construct weak occupation fields that make every support size optimal in succession. The exact law and the normal limit are known and are included for reference. Our results are the uniform estimates when the persistence varies with N , and their consequences near the boundary and for the modes.

1 Introduction

A symmetric persistent chain provides a simple model in which the most likely occupation vector changes from a vertex to the interior as the persistence decreases. The comparison involves the probability of one specific vector, rather than the aggregate probability of a face. It is especially sensitive to coordinates that are small relative to the total number of visits. This paper studies those crossing locations for a fixed number of states, including count vectors approaching a vertex at arbitrary rates.

The model is a multistate version of the Galton-board process in Kagey's Problem 131 [1]. The binary case, including the comparison of free and periodic boundaries, is treated in a companion paper [2]. This paper does not depend on it, and we prove every estimate we use. Geometrically, the occupation vectors index the bins of a simplex board with fixed global directions and uniform switching; a board with a different rule at its walls gives a different process.

Our main estimates address three regimes. On a fixed simplex face, the balanced crossing lies in the common window $Nu = \log N - \frac{1}{2} \log \log N + O(1)$, with explicit constants depending on its support size. A relative Bessel bound uniform over every positive composition resolves the transition when additional coordinates have size $N/(\log N)^2$. Finally, if one count is b and the remaining counts have total $A = o(b)$, a single polynomial equation gives the crossing with relative

error $O_\ell(\sqrt{(A+1)/b})$, uniformly over their individual sizes. This holds even in a regime where the corresponding approximation to the probabilities is not uniform. For the modes, we find the volumes of the spatial superlevel sets and give an explicit weak field under which every support size occurs in turn.

Relation to established results. The law belongs to the theory of Markov multinomial distributions [3, 4], and the exact finite sum and the normal limit below are included for reference. We cite the Wang–Yang article through Wang and Tang, since we have not been able to obtain it. Brydges, van der Hofstad and König [5] give occupation densities for continuous-time chains on the visited faces of the simplex. The square-root quantity that appears below is also classical: if Q is the continuous-time generator of the complete graph with off-diagonal entries one, then its symmetric occupation rate is

$$I_Q(\alpha) = -\langle \sqrt{\alpha}, Q\sqrt{\alpha} \rangle = d - \left(\sum_i \sqrt{\alpha_i} \right)^2.$$

Hence the exponent $(\sum_i \sqrt{\alpha_i})^2 - 1$ is the difference between the rate at a vertex and the rate at α . The Bessel expansions we use are standard [6], and we prove all the discrete comparisons and root inversions. What is new here is that these estimates hold uniformly when the persistence depends on N , and the consequences of this near the boundary and for the modes.

Organization and scope. Section 2 gives the exact law, elementary estimates, and fixed-persistence covariance and normal limit. Section 3 proves the face crossings and balancing theorem. Sections 4 and 5 treat boundary coordinates and the uniform near-vertex law. Section 6 treats superlevel sets and weak fields. We do not determine every global mode at finite N , and we do not allow the number of states to grow. Example 3.8 shows that at finite N the global mode need not be a vertex or a full center.

2 The model and exact formulas

Fix $d \geq 2$ and let $(X_j)_{j \geq 1}$ be a Markov chain on $\{1, \dots, d\}$. The initial state is uniform. At each subsequent step the chain repeats its state with probability $p \in [0, 1]$ and moves to each other state with probability $r = (1-p)/(d-1)$. Put $q = 1-p$ and $\sigma = p-r = (dp-1)/(d-1)$. Let $K_N = (K_{N,1}, \dots, K_{N,d})$ be the occupation vector over N visits, and write $P_N(k) = \mathbb{P}(K_N = k)$. For a word w of length $N \geq 1$ with j changes, its probability is $d^{-1}p^{N-1-j}r^j$. The vertex probability is $V_N = p^{N-1}/d$. For $p > 0$ put $u = r/p$; for a positive vector $n = (n_1, \dots, n_k)$ with total N , write

$$S_n(u) = \frac{P_N(n_1, \dots, n_k, 0, \dots, 0)}{V_N} = \sum_j W_n(j) u^j,$$

where $W_n(j)$ counts words on the specified k states with these occupations and j changes. The ratio is independent of the ambient number d once the positive coordinates are fixed. Set $P_0(0, \dots, 0) = 1$ separately.

Proposition 2.1 (*d directions*). *Let $P_N(k_1, \dots, k_d)$ be the probability that direction i is taken k_i times. Put $\sigma = p - \frac{q}{d-1} = \frac{dp-1}{d-1}$ and $\mathcal{R} = \sum_{i=1}^d \frac{x_i t}{1-\sigma x_i t}$. Then*

$$\sum_{N \geq 0} \sum_{k_1 + \dots + k_d = N} P_N(k_1, \dots, k_d) x_1^{k_1} \dots x_d^{k_d} t^N = 1 + \frac{\mathcal{R}/d}{1 - \frac{q}{d-1} \mathcal{R}}.$$

Proof. Let F_i be the generating function of words ending in state i , and $S = \sum_i F_i$. Conditioning on the last bounce, $F_i = x_i t \left(\frac{1}{d} + p F_i + \frac{q}{d-1} (S - F_i) \right)$, that is $F_i(1 - \sigma x_i t) = x_i t \left(\frac{1}{d} + \frac{q}{d-1} S \right)$. Summing over i gives $S = \mathcal{R} \left(\frac{1}{d} + \frac{q}{d-1} S \right)$, and the generating function is $1 + S$. $\square$

For $d = 2$, setting $x_1 = x$ and $x_2 = 1$ gives the binary generating function studied in the companion [2].

Proposition 2.2 (finite bin probabilities in d directions). *Put $r = q/(d-1)$ and $\sigma = p - r$. For $N \geq 1$, let $k = (k_1, \dots, k_d)$ have nonnegative integer coordinates summing to N , and put $H = \{i : k_i > 0\}$. Sum over $m_i = 0$ for $i \notin H$ and $1 \leq m_i \leq k_i$ for $i \in H$, and set $M = \sum_i m_i$. Then*

$$P_N(k) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in H} \binom{k_i - 1}{m_i - 1}.$$

All exponents are nonnegative, and $0^0 = 1$. Set $P_0(0, \dots, 0) = 1$ separately.

Proof. With $z_i = x_i t$, Proposition 2.1 becomes the formal power series

$$1 + \frac{1}{d} \sum_{M \geq 1} r^{M-1} \left(\sum_i \frac{z_i}{1 - \sigma z_i} \right)^M.$$

Expand by the multinomial theorem and use, for $k \geq m \geq 1$,

$$[z^k] \left(\frac{z}{1 - \sigma z} \right)^m = \binom{k-1}{m-1} \sigma^{k-m}.$$

For $m = 0$ only $k = 0$ contributes. Multiplying the coefficients proves the formula. $\square$

At $d = 3$ this is a triple sum, with the support convention covering edge bins. At $p = 1/d$, $\sigma = 0$ and only $M = N$ survives, giving the ordinary multinomial law. At $p = 1$, only $M = 1$ survives, giving mass $1/d$ at each Ne_i . The formula also holds at $p = 0$. For $\sigma < 0$ its terms can cancel, so for numerical work we use dynamic programming or exact arithmetic. This model assumes fixed global directions and uniform switching to the other directions, and a board with a different rule need not have this law.

2.1 Elementary bounds used below

For $x \geq 0$, the modified Bessel function used here is

$$I_1(x) = \sum_{j \geq 0} \frac{(x/2)^{2j+1}}{j!(j+1)!}.$$

Its coefficients give $I_1(x) \leq \sinh x \leq e^x$. We also use the standard expansion [6, Eq. 10.40.1]

$$I_1(x) = \frac{e^x}{\sqrt{2\pi x}} \left(1 - \frac{3}{8x} + O(x^{-2}) \right), \quad x \rightarrow \infty.$$

The Laguerre–Bessel comparison needed near the boundary is proved in Lemma 4.1.

Lemma 2.3 (clipped product deficit). *For any finite list of nonnegative real numbers $t_1, \dots, t_m$, with $(x)_+ = \max(x, 0)$,*

$$0 \leq 1 - \prod_{j=1}^m (1 - t_j)_+ \leq \sum_{j=1}^m t_j.$$

Proof. If some $t_j \geq 1$, the product is zero and the sum is at least one. Otherwise all factors lie in $[0, 1]$ and induction follows from $1 - (1 - a)(1 - b) = a + b - ab \leq a + b$. The empty product gives equality. $\square$

2.2 Covariance and the fixed-persistence normal limit

We record these formulas for comparison with the later sections. Note that the normal limit at fixed p says nothing about individual occupation probabilities when p depends on N .

Remark 2.4 (covariance and the tetrahedral case). For the board with d directions of Proposition 2.1 let K_a be the number of bounces in direction a . The direction chain has transition matrix $\sigma I + (1 - \sigma)\frac{1}{d}J$ with J the all-ones matrix, so the indicator of direction a at time i and of direction b at time $i + m$ have covariance $\sigma^m \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2} \right)$. Summing over the two time indices and evaluating the finite geometric sum gives

$$\text{Cov}(K_a, K_b) = \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2} \right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2} \right), \quad \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

These formulas hold for $p < 1$; at $p = 1$ the covariance is $N^2(\delta_{ab}/d - 1/d^2)$. The asymptotic multiplier of the multinomial covariance is $(d - 2 + dp)/(d(1 - p))$, which is $(1 + 3p)/(3(1 - p))$ for the tetrahedral board, $d = 3$.

Theorem 2.5 (simplex normal limit). *Let $d \geq 3$ and $0 \leq p < 1$, or $d = 2$ and $0 < p < 1$. Set*

$$\pi = (1/d, \dots, 1/d), \quad C = \frac{1}{d}I - \frac{1}{d^2}J, \quad c = \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

Then $(K_N - N\pi)/\sqrt{N}$ converges in distribution to $\mathcal{N}_d(0, cC)$. This Gaussian is supported on the sum-zero hyperplane and is nondegenerate there.

Proof. Let $\Pi = \mathbf{1}\pi$ and $T = \sigma I + (1 - \sigma)\Pi$. Fix a real vector a and put $f_i = a_i - d^{-1} \sum_j a_j$, so $\pi f = 0$ and $Tf = \sigma f$. For $D(t) = \text{diag}(e^{itf_i})$ and $B(t) = TD(t)$, stationarity gives

$$\mathbb{E} \exp \left(it \sum_{j=1}^N f(X_j) \right) = \pi D(t) B(t)^{N-1} \mathbf{1}.$$

At zero, $B(0) = T$ has the simple eigenvalue 1 and all other eigenvalues equal σ , with $|\sigma| < 1$. Analytic perturbation gives an eigenvalue $\lambda(t)$ and right eigenvector $h(t)$ with $\lambda(0) = 1$, $h(0) = \mathbf{1}$ and $\pi h(t) = 1$. The complementary spectral part is uniformly exponentially small for sufficiently small t : choose a circle of radius $\eta < 1$ enclosing the complementary spectrum but not $\lambda(t)$ and use the resolvent formula for matrix powers. Thus the characteristic function equals $b(t)\lambda(t)^{N-1} + O(\eta^N)$ with $b(t) \rightarrow 1$.

Differentiate $Bh = \lambda h$ at zero. Since $B'(0) = iT \text{diag}(f)$,

$$\lambda'(0) = 0, \quad (I - T)h'(0) = i\sigma f, \quad h'(0) = \frac{i\sigma}{1 - \sigma} f.$$

Differentiating again and multiplying by π gives

$$\lambda''(0) = \pi B''(0) \mathbf{1} + 2\pi B'(0) h'(0) = -\frac{1 + \sigma}{1 - \sigma} \pi(f^2) = -c \pi(f^2).$$

Hence $\lambda(t) = 1 - c\pi(f^2)t^2/2 + O(t^3)$. Substitute $t = s/\sqrt{N}$ to obtain the scalar normal characteristic function with variance $c\pi(f^2) = ca^\top Ca$. Lévy's theorem and the Cramér–Wold device give the result. $\square$

For $d = 3$ the first two centered coordinates have limiting covariance

$$\frac{1+3p}{27(1-p)} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix};$$

the third is minus their sum. Unlike the binary case, $p = 0$ is included for $d \geq 3$. At $d = 2, p = 0$ bounded alternation gives the zero limit after division by $\sqrt{N}$. At $p = 1$ the centered sequence on this scale is not tight.

3 Face-center thresholds on a simplex

We retain the symmetric d -direction model of Proposition 2.1: the initial direction is uniform, a repeat has probability p , and each specified change has probability $r = (1-p)/(d-1)$. Other ways of modeling a tetrahedral board ($d = 3$) give different laws, which we do not treat. For $2 \leq k \leq d$ and $N \geq k$, let $C_{N,k}(p)$ denote the probability of one *balanced face bin*: its k positive coordinates differ by at most one, and its other coordinates vanish. All such bins have the same probability. Let $V_N(p) = p^{N-1}/d$ be the probability of one vertex bin, and put

$$u = \frac{1-p}{(d-1)p}, \quad a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k.$$

The ratio $C_{N,k}/V_N$, viewed as a function of u , depends on k and N but not on the ambient number d of directions.

Theorem 3.1 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$, there is exactly one $p = p_{N,k}^{(d)} \in (0, 1)$ for which $C_{N,k}(p) = V_N(p)$. The face center is more likely below this persistence and less likely above it. The sequence $p_{N,k}^{(d)}$ is strictly increasing in N , and*

$$p_{k,k}^{(d)} = \frac{1}{1 + (d-1)(k!)^{-1/(k-1)}}.$$

Writing $L = \log N$ and $u_{N,k} = (1 - p_{N,k}^{(d)})/((d-1)p_{N,k}^{(d)})$, for fixed k we have

$$Nu_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O_k\left(\frac{(\log L)^2}{L^2}\right). \quad (1)$$

Consequently, $N(1 - p_{N,k}^{(d)})$ equals $(d-1)$ times the right-hand side of (1), with the same error order for fixed d .

In particular, for the full center $k = d$,

$$N(1 - p_{N,d}^{(d)}) = (d-1) \log N - \frac{d-1}{2} \log \log N + c_d + o(1), \quad c_d = \frac{d-1}{2} \log(4\pi) - \frac{2d+1}{2} \log d.$$

For $d = 2$, $a_2 = \frac{1}{2} \log(\pi/8)$, and the expansion agrees with the binary one in [2]. The proof below does not use the binary case and treats every face dimension at once.

Theorem 3.2 (common critical window). *Fix d and put*

$$u_N(t) = \frac{\log N - \frac{1}{2} \log \log N + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)}.$$

Uniformly for t in any fixed compact interval, for each $2 \leq k \leq d$,

$$\frac{C_{N,k}(p_N(t))}{V_N(p_N(t))} \longrightarrow \exp((k-1)(t-a_k)). \quad (2)$$

Moreover $a_2 > a_3 > \dots$. Thus the face centers cross the vertex height at distinct constant offsets within a common window of persistence width $O(1/N)$.

We first prove the exact assertions, and then prove a uniform local estimate from which both asymptotic theorems follow.

Proof of the exact assertions in Theorem 3.1. For a positive count vector $n = (n_1, \dots, n_k)$, let $W_n(j)$ be the number of words with those counts and exactly j changes. Its ratio to a vertex probability is

$$S_n(u) = \sum_j W_n(j) u^j.$$

To count $W_n(j)$, choose a proper word of $j+1$ run colors, meaning that successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. Thus $W_n(j)$ is a sum of products of these binomial coefficients. Every coefficient is nonnegative; the first nonzero one is $W_n(k-1) = k!$. Hence S_n is strictly increasing from zero to infinity on $u \geq 0$ and crosses one exactly once. At $p = 0$, cycling through the k colors and truncating after N positions gives a word with balanced counts and no successive repeat, so its probability is positive while the vertex probability is zero. At $p = 1$ the reverse strict inequality holds.

Choose the balanced count vectors as N increases by incrementing one of their minimum coordinates. Each coefficient $W_n(j)$ is then nondecreasing. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

Indeed, exactly one run color appears twice; for each such color there are $(k+1)!/2 - k! = k!(k-1)/2$ proper arrangements of the multiset of run colors. This coefficient increases strictly with N , so the positive root decreases strictly with N . The substitution $p = 1/(1 + (d-1)u)$ proves the claimed monotonicity. If $N = k$, every coordinate equals one and $S_n(u) = k!u^{k-1}$, giving the initial value. Note also that every root satisfies $u \leq (k!)^{-1/(k-1)} < 1$, so all crossings lie in the positive refresh regime used below. $\square$

Lemma 3.3 (positive integral and discrete comparison). *For fixed $k \geq 2$, define*

$$F_k(x) = \sum_{m_1, \dots, m_k \geq 1} M! \prod_{i=1}^k \frac{x^{m_i}}{m_i!(m_i-1)!} = \int_0^\infty e^{-t} [\sqrt{xt} I_1(2\sqrt{xt})]^k dt, \quad M = \sum_i m_i. \quad (3)$$

Let n be a balanced positive count vector of total N , put $h = N/k$, and, for $0 < u < 1$, put $x = hu/(1-u)$. With $\mathcal{D} = x d/dx$,

$$0 \leq 1 - \frac{S_n(u)}{(1-u)^{N-1} h^{1-k} F_k(x)/x} \leq \frac{\mathcal{D}^2 F_k(x)}{h F_k(x)} = O_k\left(\frac{(1+x)^2}{h}\right). \quad (4)$$

In particular the relative error is $O_k((\log N)^2/N)$ when $x = O(\log N)$.

Proof. The equality in (3) follows from the Bessel power series and Tonelli's theorem, using $M! = \int_0^\infty e^{-t} t^M dt$. The integral is finite for every positive x : the bound $I_1(y) \leq e^y$ for $y \geq 0$ reduces its tail to a polynomial times $\exp(-t + 2k\sqrt{xt})$. Thus the positive series and all its termwise derivatives converge on compact subsets of $x > 0$.

Set $\sigma = p - r = p(1 - u)$ and $v = r/\sigma = u/(1 - u)$. Proposition 2.2, restricted to the positive coordinates, gives

$$S_n(u) = (1 - u)^{N-1} \sum_{m_i \geq 1} M! v^{M-1} \prod_i \frac{\binom{n_i-1}{m_i-1}}{m_i!}. \quad (5)$$

Define

$$A_{i,m} = \frac{(m-1)!}{h^{m-1}} \binom{n_i-1}{m-1} = \prod_{j=1}^{m-1} \left(1 - \frac{h - n_i + j}{h}\right)_+.$$

Since $h - n_i > -1$, these are products of factors in $[0, 1]$, including beyond their finite support. The elementary product-deficit bound in Lemma 2.3 gives

$$0 \leq 1 - \prod_i A_{i,m_i} \leq \frac{1}{h} \sum_i \frac{m_i(m_i + 1)}{2} \leq \frac{M^2}{h}.$$

Replacing the binomial factors in (5) by $h^{m_i-1}/(m_i-1)!$ gives the denominator in (4). Summing the deficit bound proves its first inequality and the displayed explicit upper bound.

To bound the quotient of derivatives, for $x \geq 1$ use $M^2 \leq 3x^2(1+1/x)^M$. Indeed, $\log(1+1/x) \geq 1/(2x)$ and $\sup_{y \geq 0} y^2 e^{-y/2} = 16/e^2 < 3$. Positivity of the coefficients gives

$$\frac{\mathcal{D}^2 F_k(x)}{F_k(x)} \leq 3x^2 \frac{F_k(x+1)}{F_k(x)} = O_k(x^2),$$

where the last step follows from Lemma 3.4 below. For $0 < x \leq 1$, the quotient extends continuously to $x = 0$ with value k^2 , since $F_k(x) = k!x^k + O_k(x^{k+1})$. This proves the bound. $\square$

Lemma 3.4 (Laplace expansion). *For each fixed $k \geq 2$, as $x \rightarrow \infty$,*

$$F_k(x) = \frac{k^{k/2+1}}{(4\pi)^{(k-1)/2}} x^{(k+1)/2} e^{k^2 x} \left(1 - \frac{k-1}{8kx} + O_k(x^{-2})\right). \quad (6)$$

Proof. In (3) set $w = \sqrt{t}$ and $a = k\sqrt{x}$. The exponent becomes $-w^2 + 2k\sqrt{x}w = k^2x - (w-a)^2$. First restrict to $|w-a| \leq \sqrt{x}/2$. There $2\sqrt{x}w$ is bounded below by $(2k-1)x$, so the standard Bessel expansion [6, Eq. 10.40.1] applies uniformly:

$$[\sqrt{x}w I_1(2\sqrt{x}w)]^k = \frac{x^{k/4} w^{k/2} e^{2k\sqrt{x}w}}{(4\pi)^{k/2}} \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2})\right).$$

On the complementary set, the elementary exponential bound for I_1 dominates the integrand by $2x^{k/2} w^{k+1} e^{k^2x - (w-a)^2}$. Gaussian tail estimates therefore show that its integral is at most $e^{k^2x - cx}$ times a fixed power of x , for some $c > 0$. It is exponentially smaller than the main term.

Put $\alpha = k/2 + 1$. Apart from that negligible tail, the integral is

$$\frac{2x^{k/4} e^{k^2x}}{(4\pi)^{k/2}} \int_{|w-a| \leq \sqrt{x}/2} e^{-(w-a)^2} w^\alpha \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2})\right) dw.$$

Taylor-expand w^α about a . The odd terms integrate to zero on this symmetric interval, the quadratic term contributes $\alpha(\alpha-1)/(4a^2)$ relatively, and the fourth-order remainder is $O_k(a^{-4})$: the interval stays a fixed positive fraction of a away from zero and all Gaussian moments are finite. Extending the centered Gaussian integrals to the real line changes them exponentially little. For the term involving $w^{\alpha-1}/\sqrt{x}$, keeping its constant term suffices, because its quadratic correction has relative order x^{-2} . Since $\int e^{-y^2} dy = \sqrt{\pi}$ and $\int y^2 e^{-y^2} dy = \sqrt{\pi}/2$, the total relative correction is

$$\frac{\alpha(\alpha-1)}{4k^2x} - \frac{3}{16x} = -\frac{k-1}{8kx}.$$

The leading constant simplifies to the one in (6), completing the proof. $\square$

Proof of the asymptotic assertions in Theorems 3.1 and 3.2. Set $s = k-1$, $z = Nu$, and $D_k = k^{k+1/2}/(4\pi)^{(k-1)/2}$. The preceding lemmas imply, uniformly on $c \log N \leq z \leq C \log N$ for fixed $0 < c < C$,

$$S_n(z/N) = D_k \frac{z^{s/2} e^{sz}}{N^s} \left(1 - \frac{s}{8z} + O_{k,c,C}(z^{-2} + z^2/N) \right). \quad (7)$$

Indeed, $x = z/(k(1-u))$, and

$$k^2x + (N-1)\log(1-u) - sz = O_k(z^2/N).$$

The prefactor changes by $1 + O_k(z/N)$ on replacing x by z/k ; rounding of the balanced coordinates is covered by (4).

At $z = (1-\varepsilon)\log N$ the ratio in (7) tends to zero, whereas at $z = (1+\varepsilon)\log N$ it tends to infinity. The unique root therefore has $z/\log N \rightarrow 1$. Taking logarithms at the root and dividing by s gives

$$z + \frac{1}{2} \log z = L + a_k + \frac{1}{8z} + O_k(z^{-2} + z^2/N).$$

For an explicit inversion, put

$$d_L = -\frac{1}{2} \log L + a_k, \quad e_L = \frac{1}{8} - \frac{1}{2} d_L, \quad z_* = L + d_L + e_L/L.$$

Taylor expansion gives $z_* + \frac{1}{2} \log z_* - 1/(8z_*) = L + a_k + O_k((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2} \log z - 1/(8z)$ is at least one for $z > 0$. The preceding root equation, with $z \sim L$, therefore gives $z - z_* = O_k((\log L)^2/L^2)$, proving (1). The conversion to physical persistence uses $N(1-p) = (d-1)z/(1+(d-1)z/N) = (d-1)z + O_d(z^2/N)$. Substituting $z = L - \frac{1}{2} \log L + t$ in (7) proves (2), uniformly on compact t intervals.

For the ordering of constants, extend $g(x) = (x+1/2) \log x/(x-1)$ to $x \geq 2$. The sign of $g'(x)$ is the sign of $H(x) = 2x^2 - x - 1 - 3x \log x$. Here $H(2) = 5 - 6 \log 2 > 0$, and $H'(x) = 4x - 4 - 3 \log x > 0$ for $x \geq 2$, since its derivative is $4 - 3/x > 0$ and $H'(2) > 0$. Thus g increases strictly, so a_k decreases strictly. $\square$

3.1 Thresholds throughout the interior of a face

The same argument gives a threshold at every point in the interior of a face, not just at its center. The constants below are evaluated at the actual fractions $\alpha_i = n_i/N$, since we do not assume that the count vector converges at any rate.

Theorem 3.5 (interior composition thresholds). *Fix $k \geq 2$ and $\varepsilon > 0$. Let $n_1 + \dots + n_k = N$, with $\alpha_i = n_i/N \geq \varepsilon$, and define*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $c \log N \leq z \leq C \log N$, with fixed $0 < c < C$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} (1 + O_{k,\varepsilon,C}(z^{-1} + z^2/N)). \quad (8)$$

Let $u_n > 0$ be the unique root of $S_n(u) = 1$. Writing $L = \log N$,

$$Nu_n = \frac{s}{g}L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (9)$$

These estimates are uniform on every compact subset of the relative interior of the face. For a fixed target α and integer counts $n_i = N\alpha_i + O(1)$, the constants may equivalently be evaluated at that target, without changing the error order.

Proof. The proper-run polynomial used above has nonnegative coefficients and first coefficient $k!u^{k-1}$ for every positive count vector, which gives the unique positive root. To obtain the approximation, put $v = u/(1-u)$ and $y = Nv$, and introduce

$$G_\alpha(y) = \int_0^\infty e^{-t} \prod_i [\sqrt{\alpha_i y t} I_1(2\sqrt{\alpha_i y t})] dt.$$

Replacing each binomial coefficient in (5) by $n_i^{m_i-1}/(m_i-1)!$ gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v \prod_i n_i} G_\alpha(y) (1 + O_{k,\varepsilon}((1+y)^2/N)). \quad (*)$$

To see that the error is uniform, note that the falling-factorial deficit is bounded by $\sum_i m_i(m_i - 1)/(2n_i) \leq M^2/(2\varepsilon N)$. The positive-series argument in Lemma 3.3 bounds the weighted second moment of M by a constant times $(1+y)^2$. Uniformity follows from the uniform saddle expansion below, which bounds $G_\alpha(y+1)/G_\alpha(y)$ on the compact set $\{\alpha_i \geq \varepsilon, \sum_i \alpha_i = 1\}$; for bounded y , continuity and the positive leading coefficient give the same conclusion.

Set $w = \sqrt{t}$. The exponential factor of the integrand is

$$\exp(-w^2 + 2A\sqrt{y}w) = \exp(A^2y - (w - A\sqrt{y})^2).$$

Every α_i stays bounded away from zero, so the Bessel expansions, Gaussian tail estimates, and Taylor remainders in Lemma 3.4 are uniform here. Keeping their leading terms yields

$$G_\alpha(y) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{1/4}}{(4\pi)^{s/2}} y^{(k+1)/2} e^{A^2y} (1 + O_{k,\varepsilon}(y^{-1})).$$

Substitute this into (*) and use $v \prod_i n_i = yN^{k-1} \prod_i \alpha_i$. For $u = z/N$ in the stated range, $y = z + O(z^2/N)$ and $(N-1) \log(1-u) = -z + O(z^2/N)$. This proves (8) with the specified constant.

On the compact set of compositions, both g and $D(\alpha)$ have positive lower and finite upper bounds. Testing the ratio at suitable constant multiples of $\log N$ brackets the unique root uniformly. Taking logarithms there gives

$$gz + \frac{s}{2} \log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N).$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$; one substitution then gives (9). The derivative of its left-hand side is at least g , bounded uniformly away from zero, so the stated error is preserved on inversion. Finally, the functions of α appearing in the formula have bounded derivatives on a slightly smaller compact set; changing α by $O(1/N)$ changes the expression by $O(L/N)$, which is absorbed by its error term. $\square$

At a balanced composition, $A = \sqrt{k}$, $g = k - 1$, and $D(\alpha) = D_k$, recovering the preceding center formula through its constant term. For $k = 2$, $g = 2\sqrt{\alpha_1\alpha_2}$, recovering the binary bulk scale. Cauchy–Schwarz gives $g \leq k - 1$, with equality only at the balanced composition, so the leading threshold coefficient s/g is minimized exactly at the center of each face.

3.2 Balanced bins and global modes

Theorem 3.6 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , probability is maximized precisely by the balanced count vectors. More precisely, transferring one unit from a coordinate b to another coordinate a , where $b \geq a + 2$ and $a \geq 1$, strictly increases the bin probability.*

Proof. At $p = 1/d$ this follows from the ordinary multinomial formula. For $1/d < p < 1$, put $\sigma = p - r > 0$ and $v = r/\sigma > 0$. The positive refresh sum has the integral form

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}. \quad (10)$$

For each $y > 0$, the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. This is a case of the fact that the binomial transform preserves log-concavity, and we give a direct proof. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the expectation of the strictly decreasing sequence $a_{i+1}/a_i = y/(i+2)$ under weights proportional to $\binom{j}{i} a_i$. When j increases to $j+1$, the likelihood ratio of the new and old unnormalized weights on the old support is $(j+1)/(j+1-i)$, increasing in i ; additionally a strictly positive weight appears at $i = j+1$. Thus these weights shift strictly toward larger indices, so the expectation of the strictly decreasing sequence decreases strictly. To check this last step, cross-multiply the two expectations and pair the terms with indices $i < \ell$. Hence $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j , proving the asserted strict log-concavity.

It follows that $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$ whenever $b \geq a + 2$. Apply this inequality pointwise for $y = vt > 0$ in (10) and integrate. Repeated transfers give the conclusion. $\square$

Corollary 3.7 (global modes in the critical window). *Fix d . In the window of Theorem 3.2, if $t < a_d$, then for all sufficiently large N the only global maxima are the d vertices. If $t > a_d$, then the only global maxima are the fully supported balanced bins. These assertions are uniform on*

compact sets of t separated from a_d . At the exact full-center/vertex crossing, for all sufficiently large N the global maxima are exactly the vertices and the fully supported balanced bins, with both classes present.

Proof. Theorem 3.6 reduces the possible modes to balanced face bins of support sizes $1, \dots, d$. For $2 \leq k \leq d$ their limiting log ratios to a vertex are $(k-1)(t-a_k)$. If $t < a_d$, these are all negative. If $t > a_d$, the full-center ratio exceeds the vertex ratio, and for $2 \leq k < d$ its limiting log ratio exceeds the k -face one by

$$(d-k)(t-a_d) + (k-1)(a_k-a_d) > 0.$$

Uniformity in Theorem 3.2 gives both assertions. At the exact full-center crossing, its window parameter tends to a_d by Theorem 3.1; every proper face-center ratio then has a limit strictly less than one. The same reduction proves the final claim. $\square$

Example 3.8 (an intermediate face can be a finite-size mode). Corollary 3.7 does not hold for every N . Take $d = 3$, $N = 4$, and $p = 10/17$, so $u = 7/20$. For an edge center and a full center respectively,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

By Theorem 3.6, the three edge centers are the only global modes. So at finite N the modes can lie on an intermediate face.

4 A uniform simplex estimate and a boundary crossover

The compact-interior assumption in Theorem 3.5 excludes bins approaching a lower-dimensional face. We now remove that assumption from the Bessel estimate and study the transition in which some positive coordinates become small. The transition below needs at least two coordinates of order N . With only one such coordinate the persistence scale is different, and we treat that case at the end of this section and in Section 5.

Put

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

As before, $S_n(u)$ is the probability of the positive count vector n divided by the probability of one vertex, and $u = r/p$. None of the statements depend on d once the support is fixed.

Lemma 4.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (11)$$

Proof. The coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\prod_{h=1}^j (1-h/a)_+$. Its deficit from one lies between zero and $j(j+1)/(2a)$, including for $j \geq a$. Consequently

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw).$$

$\square$

Theorem 4.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1 - u)$. Define*

$$\mathcal{Q}_n(u) = (1 - u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i v t) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2}v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2}Nv^2. \quad (12)$$

Thus this approximation has relative error $O_k((\log N)^2/N)$ whenever $u = O(\log N/N)$, uniformly over every positive composition of N .

Proof. The positive refresh formula gives

$$S_n(u) = \frac{(1 - u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt.$$

Applying Lemma 4.1 to each factor, and then the product-deficit inequality, bounds the relative deficit by $(kv/2)\mathbb{E}[T]$, where T has density proportional to $e^{-t} t^k \prod_i \Phi(n_i v t)$. For the probability weights proportional to $x^j/[j!(j+1)!]$, the identity $\mathbb{E}[J(J+1)] = x$ implies

$$x \frac{\Phi'(x)}{\Phi(x)} = \mathbb{E}[J] \leq \sqrt{x}.$$

Integration by parts therefore gives

$$\mathbb{E}[T] = k + 1 + \sum_i \mathbb{E} \left[\frac{n_i v T \Phi'(n_i v T)}{\Phi(n_i v T)} \right] \leq k + 1 + \sqrt{v} \left(\sum_i \sqrt{n_i} \right) \sqrt{\mathbb{E}[T]}.$$

The boundary terms vanish: near zero the density is $O(t^k)$, while $\Phi(x) \leq e^{2\sqrt{x}}$ bounds the tail by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$. This also proves finiteness of the integral and of every moment used above. Applying $2ab \leq a^2 + b^2$ to the final term yields $\mathbb{E}[T] \leq 2(k+1) + v(\sum_i \sqrt{n_i})^2$. The claimed estimates follow, the last one by Cauchy–Schwarz. $\square$

4.1 Coordinates of order $N/(\log N)^2$

Fix integers $r \geq 2$ and $\ell \geq 1$, and put $k = r + \ell$. Consider r large counts $n_1, \dots, n_r$ and ℓ further positive counts $a_1, \dots, a_\ell$. Write

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j below always refer to these actual counts. Define

$$A = \sum_{i=1}^r \sqrt{\beta_i}, \quad g = A^2 - 1, \quad K = r + \ell - 1, \quad H = \frac{r-1}{2} + 2\ell, \quad c = \frac{K}{g},$$

and

$$D_r(\beta) = \frac{A^{r/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(r-1)/2}}.$$

Theorem 4.3 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. Uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$, where $0 < c_0 < C_0 < \infty$ are fixed,*

$$S_{(n,a)}(z/N) = D_r(\beta) A^{2\ell} \frac{z^H e^{gz}}{N^K} \prod_{j=1}^{\ell} \Phi\left(\frac{A^2 a_j z^2}{N}\right) \left(1 + O\left(L^{-1} + \frac{L^2}{N}\right)\right). \quad (13)$$

The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ consequently satisfies

$$\begin{aligned} N u_{n,a} = cL - \frac{H}{g} \log L - \frac{1}{g} \left\{ \log(D_r(\beta) A^{2\ell}) + H \log c + \sum_{j=1}^{\ell} \log \Phi(A^2 c^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (14)$$

The constants in the errors depend only on the displayed fixed parameters.

Proof. Let $v = u/(1-u)$ and $Y = Mv$. By Theorem 4.2, with relative error $O_k(v + Nv^2)$, the exact ratio equals

$$\frac{(1-u)^{N-1} v^{\ell-1}}{\prod_{i=1}^r n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^r \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^{\ell} \Phi(a_j v t) dt. \quad (15)$$

In the stated regime $Y \asymp L$ and all $a_j v Y$ are bounded. Set $w = \sqrt{t}$. The exponential from the r large-coordinate factors is $\exp(A^2 Y - (w - A\sqrt{Y})^2)$. On $|w - A\sqrt{Y}| \leq \sqrt{Y}/2$, the large-coordinate Bessel arguments are bounded below by a positive constant times Y . Their usual large-argument expansions are uniform, with relative error $O(1/Y)$. The remaining smooth factor, apart from constants and powers of Y , is

$$w^{r/2+1+2\ell} \prod_j \Phi(a_j v w^2).$$

Divide this factor by its value at $w = A\sqrt{Y}$. On the same interval its second derivative is $O(1/Y)$, uniformly: $w/\sqrt{Y}$ lies in a fixed compact subset of $(0, \infty)$, the quantities $a_j v Y$ lie in a fixed compact interval, and $\Phi \geq 1$ there. Taylor's theorem, followed by integration against the centered Gaussian, eliminates the linear term and gives relative error $O(1/Y)$.

The part outside this interval is exponentially small, uniformly. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ give a polynomial times

$$\exp\left(A^2 Y - (w - A\sqrt{Y})^2 + 2w \sum_j \sqrt{a_j v}\right).$$

Here $\sum_j \sqrt{a_j v} = O(Y^{-1/2})$, so completing the square leaves a Gaussian centered only $O(Y^{-1/2})$ away from $A\sqrt{Y}$. On the excluded region its integral is exponentially small relative to the main term, even after the polynomial factors are included. The same argument also justifies extending the central Gaussian integrals to the full real line.

Evaluation of the leading Gaussian integral in (15) now gives

$$\begin{aligned} S_{(n,a)}(u) = D_r(\beta) A^{2\ell} \frac{Y^H}{M^K} e^{A^2 Y + (N-1) \log(1-u)} \prod_j \Phi(A^2 a_j M v^2) \\ \times \left(1 + O(Y^{-1} + v + Nv^2)\right). \end{aligned}$$

Since $\sum_j a_j/N = O(L^{-2})$, at $u = z/N$ we have

$$Y = z + O(L^{-1} + L^2/N), \quad A^2Y + (N-1)\log(1-u) = gz + O(L^{-1} + L^2/N).$$

Replacing Y by z , M by N , and $A^2a_jMv^2$ by $A^2a_jz^2/N$ in the other factors changes the expression by a relative $O(L^{-2} + L/N)$, because the arguments of Φ stay bounded and $\Phi \geq 1$. This proves (13).

The proper-run polynomial has nonnegative coefficients, zero constant coefficient, and positive coefficient $k!$ at u^{k-1} ; hence the crossing is unique. On the compact set of permitted β , both g and $D_r(\beta)$ are bounded above and bounded away from zero. Testing (13) at fixed sufficiently small and large multiples of L brackets the root uniformly. Its logarithm gives

$$gz + H \log z = KL - \log(D_r(\beta)A^{2\ell}) - \sum_j \log \Phi(A^2\rho_j(z/L)^2) + O(L^{-1} + L^2/N).$$

First $z = cL + O(\log L)$, so the sum of logarithms on the right differs from its value at $z = cL$ by $O(\log L/L)$. Substituting this estimate into $\log z$ proves (14); inversion preserves the error because the derivative of $gz + H \log z$ is at least $g > 0$. $\square$

Corollary 4.4 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem 4.3, if all $a_j = o(N/(\log N)^2)$, the sum involving Φ in (14) is $o(1)$. Thus, at fixed actual large composition β , the rare counts do not affect the expansion through its constant term. Their number ℓ still changes both its leading and its logarithmic coefficients.*

The scale $N/(\log N)^2$ applies near a face with at least two coordinates of order N . It does not apply at a binary endpoint, where only one coordinate is large and $g = 0$; that case is treated below and in Section 5. Theorem 4.3 gives no error bound when some ρ_j is unbounded.

Lemma 4.5 (exact insertion of a singleton color). *Let n have total size M and let $\mathcal{D} = u d/du$. Then*

$$S_{(n,1)}(u) = [2u + (M-1)u^2]S_n(u) + (u - u^2)\mathcal{D}S_n(u).$$

Proof. Insert the new color into a word of length M with j changes. The two end positions contribute a factor $2u$; the j change gaps contribute ju , and the $M-1-j$ repeat gaps contribute $(M-1-j)u^2$. Removing the unique new letter reverses this construction. Summing the resulting identity over all words proves the formula. $\square$

The singleton identity gives an exact finite-size check of the factor A^2z^2/N that one very small coordinate contributes in (13).

4.2 Fixed rare counts near a vertex

With only one large coordinate, the limit is a finite polynomial instead. In this result the rare counts are fixed, and the large coordinate b , not the total count, is the asymptotic parameter.

Theorem 4.6 (a multicolor Laguerre limit at a vertex). *Fix $\ell \geq 1$ and positive integers $a_1, \dots, a_\ell$. Define*

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad Q(y) = \sum_{i < j} B'_{a_i}(y) B'_{a_j}(y) \prod_{h \neq i,j} B_{a_h}(y).$$

The sum defining Q is zero when $\ell = 1$. Uniformly on compact subsets of $y > 0$, as $b \rightarrow \infty$,

$$S_{(b,a)}\left(\sqrt{y/b}\right) = H(y) + \frac{2\sqrt{y}}{\sqrt{b}}(H'(y) + yQ(y)) + O_a(b^{-1}). \quad (16)$$

Let $y_a > 0$ be the unique root of $H(y_a) = 1$, and put $c_a = \sqrt{y_a}$. The unique endpoint-crossing odds satisfy

$$u_{b,a} = \frac{c_a}{\sqrt{b}} - \frac{1 + y_a Q(y_a)/H'(y_a)}{b} + O_a(b^{-3/2}). \quad (17)$$

For one rare color this recovers the binary fixed-edge correction $-1/b$. For at least two rare colors the coefficient of $1/b$ is strictly less than -1 .

Proof. Fix rare-run counts $m_i \in \{1, \dots, a_i\}$, put $m = \sum_i m_i$, and let q be the number of runs of the large color. The rare run lengths can be chosen in $\prod_i \binom{a_i-1}{m_i-1}$ ways. Since successive large-color runs must be separated, $q \leq m + 1$; the large run lengths have $\binom{b-1}{q-1}$ choices. At $u = \sqrt{y/b}$, configurations with these run counts contribute $O_a(b^{(q-m-1)/2})$. There are finitely many possible m_i, q , independently of b , so only $q = m + 1$ and $q = m$ affect the expansion through order $b^{-1/2}$.

If $q = m + 1$, the run word starts and ends in the large color and alternates between large and rare colors. There are $m!/\prod_i m_i!$ rare orders. Using $\binom{b-1}{m} = b^m/m! + O_a(b^{m-1})$ and summing gives $H(y)$, with an $O_a(b^{-1})$ error.

If $q = m$, there are two possibilities. One endpoint is rare and the run word otherwise alternates, giving $2m!/\prod_i m_i!$ orders. These terms sum to $2\sqrt{y} H'(y)/\sqrt{b}$. Otherwise both endpoints are large and exactly one pair of consecutive rare runs is present. The members of that pair must have different colors. For an ordered pair of colors i, j , contracting the marked adjacent pair shows that there are $(m-1)!/[(m_i-1)!(m_j-1)!\prod_{h \neq i,j} m_h!]$ orders. Summing over the two orientations and all $i < j$ gives $2y^{3/2}Q(y)/\sqrt{b}$. Terms with $q \leq m - 1$ and the next terms of the binomial expansions are $O_a(b^{-1})$, uniformly for y in a positive compact set. This proves (16).

The polynomial H has positive coefficients, tends to zero at zero, and grows to infinity, so y_a exists uniquely and $H'(y_a) > 0$. The same positivity argument gives the unique crossing for each b . First bracket it using (16) to obtain $bu_{b,a}^2 \rightarrow y_a$. The finite run expansion may also be differentiated uniformly on this compact neighborhood; its derivative tends to $H'(y_a) > 0$. Substitution of $u = c_a/\sqrt{b} + \alpha/b$ into (16) then forces $\alpha = -1 - y_a Q(y_a)/H'(y_a)$, and the uniform remainder gives (17). If $\ell \geq 2$, all terms in $Q(y)$ are positive for $y > 0$, proving the strict inequality. $\square$

If all ℓ rare colors occur once, then $H(y) = y^\ell$ and $y_a = 1$, giving the explicit specialization

$$u_{b,1,\dots,1} = b^{-1/2} - \frac{\ell+1}{2b} + O_\ell(b^{-3/2}).$$

An independent exact formula in this case is

$$S_{(b,1,\dots,1)}(u) = \ell! \sum_{q=1}^{\min(b,\ell+1)} \binom{\ell+1}{q} \binom{b-1}{q-1} u^{q+\ell-1}.$$

Indeed, order the ℓ distinct rare letters, choose the q gaps among their $\ell + 1$ end and interior gaps that contain large-color runs, and choose the positive lengths of those runs. In particular two singleton rare colors give

$$u_{b,1,1} = b^{-1/2} - \frac{3}{2}b^{-1} + O(b^{-3/2}).$$

This agrees with the exact polynomial $S_{(b,1,1)}(u) = 6u^2 + 6(b-1)u^3 + (b-1)(b-2)u^4$. Theorem 4.6 is not uniform in the rare counts. Section 5 gives the leading crossing uniformly for total rare count $A = o(b)$.

5 Uniform completion of the near-vertex regime

For the leading term of the crossing, the fixed-count hypothesis in Theorem 4.6 can be removed, and the result is uniform over all rare counts with total $o(b)$. Proposition 5.5 shows that the corresponding approximation to the probabilities is not uniform: its error can diverge while the relative error of the crossing tends to zero.

Throughout this section, $\ell \geq 1$ is fixed, $a_1, \dots, a_\ell$ are positive integers, $A = \sum_i a_i$, and

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad \kappa(y) = \frac{yH'(y)}{H(y)}.$$

Write y_a for the unique positive solution of $H(y_a) = 1$.

Lemma 5.1 (elasticity and concavity of the rare-count polynomial). *The polynomial H is convex, $\log H$ is concave on $(0, \infty)$, and κ is nondecreasing. Moreover,*

$$\frac{1}{A+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell Ay}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{Ay}\} \quad (0 < y \leq 1). \quad (18)$$

Proof. Convexity follows directly from the nonnegative coefficients. For one factor, give $J \in \{0, \dots, a-1\}$ probabilities proportional to

$$r_j = \binom{a-1}{j} \frac{y^j}{(j+1)!}.$$

The successive ratio $R(j) = (j+1)r_{j+1}/r_j = y(a-1-j)/(j+2)$, with $R(a-1) = 0$, is nonincreasing. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}R(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)R(J)]$. Therefore

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, R(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J - J')(R(J) - R(J'))] \leq 0$. Putting $M = J + 1$ and $\kappa_a = yB'_a/B_a = \mathbb{E}M$, we obtain

$$(\log B_a)'' = \frac{\text{Var } M - \mathbb{E}M}{y^2} \leq 0, \quad y\kappa'_a = \text{Var } M \geq 0.$$

This proves the concavity of $\log H$ and the monotonicity of κ .

The elementary differential equation $yB''_a + yB'_a = aB_a$ gives

$$ay = \kappa_a^2 + \text{Var } M + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } M \leq \kappa_a - 1$ yields $\kappa_a \leq 1 + \sqrt{ay}$ and $\kappa_a^2 + y\kappa_a \geq ay + 1$. For $0 < y \leq 1$, these imply $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors and applying Cauchy–Schwarz proves the elasticity bounds. Finally, $B_a(1) \geq 1$ and

$$B_a(y) \leq y\Phi(ay) \leq ye^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. Thus $H(1) \geq 1$, whereas $H(1/(A+1)) \leq (A+1)^{-\ell} e^{A/(2(A+1))} < 1$. $\square$

Lemma 5.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1 - u)$. Let G have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$T = G + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1 - u)^A \mathbb{E}[H(vT)], \quad \mathbb{E}e^{sT} = (1 - s)^{-2} \left(1 + \frac{us}{1 - s}\right)^{b-1} \quad (s < 1). \quad (19)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1 - u)]$. Whenever $s < 1$,

$$(1 - u)^A H(y) \leq S_{(b,a)}(u), \quad (20)$$

$$S_{(b,a)}(u) \leq (1 - u)^A H(y) e^{-\kappa} (1 - s)^{-2} \left(1 + \frac{us}{1 - s}\right)^{b-1}. \quad (21)$$

Proof. The exact positive-refresh integral is

$$S_{(b,a)}(u) = \frac{(1 - u)^{b+A-1}}{v} \int_0^\infty e^{-t} B_b(vt) H(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $v(1 + v)^{b-1}$. After normalization its density is a mixture of gamma densities with shape $J + 2$, where $J \sim \text{Bin}(b - 1, u)$. This is exactly the distribution of T , proving the identity and its moment generating function. Since $\mathbb{E}T = 2 + (b - 1)u$, $v\mathbb{E}T \geq bu^2 = y$. Convexity and monotonicity of H prove (20). Concavity of $\log H$ gives, for $w > 0$,

$$H(w) \leq H(y) \exp\left(\frac{\kappa}{y}(w - y)\right).$$

Insert $w = vT$ and use the moment generating function to prove (21). $\square$

Theorem 5.3 (all growing rare counts near a vertex). *For fixed ℓ , uniformly over all positive rare counts with $A = o(b)$, the unique positive crossing odds satisfy*

$$u_{b,a} = \sqrt{\frac{y_a}{b}} \left(1 + O_\ell\left(\sqrt{\frac{A+1}{b}}\right)\right), \quad \prod_i B_{a_i}(y_a) = 1. \quad (22)$$

Explicitly, put

$$K = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{A+1}{b}}, \quad \delta = 16K\varepsilon.$$

If $\delta \leq 1/2$, then

$$(1 - \delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1 + \delta)\sqrt{y_a/b}. \quad (23)$$

The constants are not optimized. No condition such as $A(\log A)^2 = o(b)$ is needed.

Proof. The proper-run polynomial has positive coefficients, zero constant coefficient, and is unbounded at infinity; its crossing is unique. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. Since $\delta \leq 1/2$ forces $\varepsilon \leq 1/(32K)$ and $y_a \leq 1$, at $u_+ = (1 + \delta)u_0$ we have $u_+ \leq \frac{3}{2}\varepsilon < 1/4$. Monotonicity of κ and the lower bound in Lemma 5.1 give

$$\log H(bu_+^2) \geq 2\kappa_0 \log(1 + \delta), \quad \frac{Au_0}{\kappa_0} \leq 2\sqrt{A/b} \leq 2\varepsilon.$$

Consequently, by (20),

$$\log S(u_+) \geq 2\kappa_0 \log(1 + \delta) - \frac{Au_+}{1 - u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon \right) > 0.$$

At $u_- = (1 - \delta)u_0$, write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(A + 1)$, $\delta \leq 1/2$, and (18) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1 - u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4K\varepsilon \leq \frac{1}{8}.$$

Using $\log(1 + x) \leq x$, $-\log(1 - s) \leq s/(1 - s)$ and $u_-, s \leq 1/4$ in (21), and dropping the nonpositive term $A \log(1 - u_-)$, gives

$$\begin{aligned} \log \frac{S(u_-)}{H(y)} &\leq -\kappa + \frac{2s}{1 - s} + \frac{bu_-s}{1 - s} \\ &= \kappa \left(\frac{1}{(1 - u_-)(1 - s)} - 1 \right) + \frac{2s}{1 - s} \\ &\leq 5\kappa(u_- + s) \leq 25K\kappa\varepsilon. \end{aligned}$$

Here $\kappa \geq \ell \geq 1$ was used in the last two inequalities. Again by monotonicity of κ ,

$$\log H(y) \leq 2\kappa \log(1 - \delta) \leq -2\kappa\delta.$$

Thus $\log S(u_-) \leq \kappa(-32K + 25K)\varepsilon < 0$. The two strict inequalities and uniqueness bracket the crossing. $\square$

Corollary 5.4 (an explicit shape law for growing rare counts). *Suppose $A \rightarrow \infty$, $A = o(b)$, and the actual proportions $\alpha_i = a_i/A$ stay in a fixed compact subset of the positive ℓ -simplex. Put $h = \sum_i \sqrt{\alpha_i}$. Then*

$$u_{b,a} \sim \frac{\ell \log A}{2h\sqrt{Ab}}.$$

More precisely, the polynomial root satisfies

$$2h\sqrt{Ay_a} = \ell \log A - \frac{\ell}{2} \log \log A + \frac{\ell}{2} \log \left(\frac{8\pi h}{\ell} \right) + \frac{3}{4} \sum_i \log \alpha_i + O_\ell \left(\frac{\log \log A}{\log A} + \frac{(\log A)^2}{A} \right).$$

Replacing $\sqrt{Ay_a}$ on the left by $\sqrt{Ab}u_{b,a}$ incurs an additional $O_\ell(\log A \sqrt{A/b})$ error.

Proof. Lemma 4.1, uniformly when $y \rightarrow 0$, gives $H(y) = y^\ell \prod_i \Phi(a_i y)(1 + O_\ell(y))$. Write $x = \sqrt{Ay}$. Bracketing first at fixed small and large multiples of $\log A$ shows that the root has $x \asymp \log A$. The standard large-argument expansion of I_1 then gives uniformly over the stated proportions

$$H(x^2/A) = \frac{x^{\ell/2} e^{2hx}}{A^\ell (4\pi)^{\ell/2} (\prod_i \alpha_i)^{3/4}} (1 + O_\ell(x^{-1} + x^2/A)).$$

Taking logarithms, substituting $x = (\ell/(2h)) \log A + O_\ell(\log \log A)$ in the logarithmic term, and using Theorem 5.3 proves the claims. The constants also depend on the fixed compact set of proportions. $\square$

Proposition 5.5 (probability approximation can fail while roots agree). *Let $a = \lfloor b/\log b \rfloor$, let y_a solve $B_a(y_a) = 1$, and set $u_0 = \sqrt{y_a/b}$. Then*

$$H(y_a) = B_a(y_a)^2 = 1, \quad S_{(b,a,a)}(u_0) \rightarrow \infty, \quad \frac{u_{b,a,a}}{u_0} \rightarrow 1.$$

Hence the crossing can be approximated uniformly even where $S(u) \sim H(bu^2)$ fails, so that approximation is not uniform over $A = o(b)$.

Proof. Retain only run words that start and end in the large color and have exactly one adjacent pair of distinct rare colors. The run enumeration in Theorem 4.6 shows that their contribution is the polynomial $2uyB'_a(y)^2$, with each term of total rare-run count m additionally multiplied by $\prod_{j=1}^{m-1} (1 - j/b)_+$. Give J probabilities proportional to $\binom{a-1}{j} y^j/j!$, the coefficients of $B'_a(y)$. Its differential equation $y(B'_a)'' + (1+y)(B'_a)' - (a-1)B'_a = 0$ implies $\mathbb{E}J^2 + y\mathbb{E}J = (a-1)y$; hence $\mathbb{E}J \leq \sqrt{ay}$ and $\mathbb{E}J^2 \leq ay$. For two independent copies, $m = J_1 + J_2 + 2$, so

$$\mathbb{E}[m(m-1)] \leq 4ay + 6\sqrt{ay} + 2.$$

The clipped-product deficit bound therefore proves

$$S_{(b,a,a)}(u_0) \geq \frac{2\kappa_a(y_a)^2}{bu_0} \left(1 - \frac{4ay_a + 6\sqrt{ay_a} + 2}{2b} \right).$$

The preceding corollary for one rare color gives $\sqrt{ay_a} \sim \frac{1}{2} \log a$, and the elasticity bounds give $\kappa_a(y_a) \sim \sqrt{ay_a}$. The last display is asymptotic to a lower bound $\sqrt{a/b} \log a \sim \sqrt{\log b}$, which diverges. The root assertion follows directly from Theorem 5.3, since $2a = o(b)$. $\square$

6 Spatial superlevel sets and weak-field face phases

The interior-composition estimate also determines the geometry of the bins whose individual probabilities exceed the vertex probability. We state it with a weak external field, since the field produces face phases that do not appear in the symmetric window.

Fix d and the symmetric transition model of Section 3. For $h = (h_1, \dots, h_d) \in \mathbb{R}^d$, define the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (24)$$

Thus the field on each step is h_i/N . This tilts the occupation law; it does not change the transition probabilities of the chain. The normalization cancels in every comparison of bins. By a modal phase we mean the face that contains the largest individual bin probabilities, not a face that carries most of the mass. Put $L = \log N$, keep

$$u_N(t) = \frac{L - \frac{1}{2} \log L + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)},$$

and write $h_* = \max_i h_i$ and $\tilde{V}_{N,h} = \max_i \tilde{P}_{N,h}(Ne_i)$. For a support $S \subseteq \{1, \dots, d\}$ of cardinality $k \geq 2$, set

$$\bar{h}_S = \frac{1}{k} \sum_{i \in S} h_i, \quad B_S(t, h) = (k-1)(t - a_k) + \bar{h}_S - h_*, \quad H_S = \{y \in \mathbb{R}^S : \sum_{i \in S} y_i = 0\}.$$

Euclidean measure on H_S is understood, and $\omega_s = \pi^{s/2}/\Gamma(1 + s/2)$ denotes the volume of its s -dimensional unit ball when $s = k-1$.

Theorem 6.1 (Gaussian superlevel geometry). *For a count vector with support exactly S , put*

$$y_i = \sqrt{L} \left(\frac{n_i}{N} - \frac{1}{k} \right), \quad i \in S.$$

Uniformly when t, h, y range over fixed compact sets, with $y \in H_S$,

$$\frac{\tilde{P}_{N,h}(n)}{\tilde{V}_{N,h}} = \exp \left(B_S(t, h) - \frac{k^2}{4} \sum_{i \in S} y_i^2 + o(1) \right). \quad (25)$$

Let $M_{N,S}(t, h)$ be the number of bins of support exactly S whose tilted probability is strictly greater than $\tilde{V}_{N,h}$. Then, uniformly for (t, h) in compact sets,

$$\frac{L^{(k-1)/2}}{N^{k-1}} M_{N,S}(t, h) \longrightarrow \frac{\omega_{k-1}}{\sqrt{k}} \left(\frac{4[B_S(t, h)]_+}{k^2} \right)^{(k-1)/2}, \quad [x]_+ = \max(x, 0). \quad (26)$$

If $B_S(t, h) < 0$, there are no such bins for all sufficiently large N . At $B_S(t, h) = 0$ the normalized count tends to zero, but we do not claim that the set is eventually empty.

Proof. With $\alpha_i = 1/k + y_i/\sqrt{L}$ and $\sum_i y_i = 0$, Taylor expansion on any fixed compact set in H_S gives

$$\left(\sum_i \sqrt{\alpha_i} \right)^2 = k - \frac{k^2}{4L} \sum_i y_i^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these expressions and $z = L - \frac{1}{2} \log L + t$ into Theorem 3.5. Since $\log D_k = -(k-1)a_k$, the logarithm of the untilted ratio is

$$(k-1)(t - a_k) - \frac{k^2}{4} \sum_i y_i^2 + o(1).$$

The logarithm of the field factor relative to the largest vertex is $h \cdot \alpha - h_* = \bar{h}_S - h_* + o(1)$, uniformly on the stated compact sets. This proves (25).

It remains to rule out superlevel bins near the boundary of the face, where the local estimate does not apply. Fix compact sets of t, h , and choose a sufficiently large constant R . Starting from any positive count vector on S , successively transfer one unit from a coordinate b to a coordinate a when $b \geq a + 2$. These transfers preserve the support and terminate at a balanced vector. By Theorem 3.6, the untilted probability strictly increases at every transfer. Each transfer changes the scaled vector y by norm $\sqrt{2L}/N$ and decreases its squared norm. If the starting vector has $\|y\| > R$, the first vector along this path with norm at most R has norm $R + o(1)$; it lies in a compact subset of the relative interior of the face. Its untilted log ratio to a vertex is at most

$$(k-1)(t - a_k) - k^2 R^2/4 + o(1).$$

This is an upper bound for the starting vector's untilted log ratio. For every composition α , $h \cdot \alpha - h_* \leq 0$. Consequently R can be chosen uniformly so that every tilted superlevel bin satisfies $\|y\| \leq R$. The same argument, with a larger R if necessary, bounds all face maximizers when comparison is made with its balanced bin.

Inside this ball the logarithms in (25) converge uniformly. For any $\delta > 0$, their positive set is sandwiched between the lattice points in balls of squared radii $4[B_S - \delta]_+/k^2$ and $4[B_S + \delta]_+/k^2$, with the inner ball interpreted as empty when $B_S \leq \delta$. The scaled count vectors form a translate of the lattice

$$\frac{\sqrt{L}}{N} \{z \in \mathbb{Z}^S : \sum_i z_i = 0\} \quad \text{in } H_S.$$

Its covolume is $\sqrt{k}(\sqrt{L}/N)^{k-1}$: the basis $e_i - e_k$, $1 \leq i < k$, has Gram determinant k . Lattice Riemann sums over balls therefore give the stated volume constant. Letting $\delta \downarrow 0$ proves (26), including $B_S = 0$. The radii stay bounded on compact parameter sets, so the same sandwich proves uniform convergence. If $B_S < 0$, the uniform profile on the confining ball is strictly below one for large N , proving emptiness. $\square$

At zero field, all supports of the same size have the same count, so the right-hand side of (26) is multiplied by $\binom{d}{k}$ when counting all bins with k positive coordinates. For $k = 2$, writing $y = (y_1, -y_1)$ gives the profile $\exp(t - a_2 - 2y_1^2)$ and the one-face count constant $\sqrt{2[t - a_2]_+}$, which agrees with the free binary window once the different centering is taken into account.

Theorem 6.2 (weak-field modal phase diagram). *Define the face scores*

$$\lambda_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, expressed relative to the untilted vertex before normalizing (24), satisfies

$$\log \max_n \left\{ e^{h \cdot n/N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \lambda_S(t, h). \quad (27)$$

For fixed (t, h) , every global mode has, for all sufficiently large N , a support maximizing the limiting score. If the maximizing support is unique, all global modes have exactly that support. On any selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

This does not identify the maximizing vectors exactly, and equal limiting scores do not imply that both supports hold modes at finite N .

If the fields are sorted $h_1 \geq \dots \geq h_d$, the score maximization reduces to the upper envelope of the d lines

$$\lambda_1(t) = h_1, \quad \lambda_k(t) = (k - 1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (28)$$

Away from ties, the winning support size is nondecreasing as t increases. If the fields are strictly ordered, the corresponding supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d - 1\}$ can be the unique winning size on an open interval for suitable fixed fields. Thus proper-face phases can persist asymptotically under a field of order $1/N$ per step.

Proof. For each fixed support, the confinement argument in the preceding proof applies also to its maximum, because the field factor is bounded above and below uniformly in N . On that compact scaled region, the profile relative to the untilted vertex is $\exp(\lambda_S - k^2\|y\|^2/4 + o(1))$. Balanced integer vectors have $y = o(1)$, so the face maximum converges to e^{λ_S} . There are finitely many supports. Taking their maximum proves (27), and the positive gap to any nonmaximizing score excludes its support for large N . If a sequence of maximizers within a selected face had scaled norm bounded away from zero, the uniform Gaussian profile would make it strictly smaller than that face's balanced bin. This proves the location statement.

Among supports of a given size k , the largest field average is the average of the k largest fields. This proves (28). For $t_1 < t_2$, a line of strictly larger slope cannot win at t_1 and lose to a line of smaller slope at t_2 : subtracting the two comparisons would give $(k_1 - k_2)(t_2 - t_1) \leq 0$. Hence the winning size is nondecreasing away from ties.

To realize a prescribed $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$. Choose $t > a_k$ sufficiently close to a_k so that $t < a_{k-1}$ if $k > 2$. The k -face score is positive; all smaller nonvertex face scores are negative, and the largest vertex score is zero. For $j > k$, the largest score is

$$(j-1)(t - a_j) - H \frac{j-k}{j}.$$

Taking H sufficiently large makes each of these strictly smaller than the k -face score. Every other support of size k includes a negative field and also loses. All comparisons are strict, so they persist on an open interval of t . Small perturbations can make the fields strictly ordered without removing the phase. $\square$

Corollary 6.3 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and choose*

$$h_i = -H(i-1)^2, \quad 1 \leq i \leq d.$$

In particular $h_i = -(i-1)^2$ is a single explicit choice valid in every fixed dimension. Put

$$c_k = (k + \tfrac{1}{2}) \log k - \frac{k-1}{2} \log(4\pi), \quad \tau_k = c_k - c_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and set $\tau_0 = -\infty$, $\tau_d = +\infty$. Then $\tau_1 < \dots < \tau_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support throughout the nonempty interval $\tau_{k-1} < t < \tau_k$. Consequently, for each fixed t in this interval and all sufficiently large N , every global mode of the tilted law has exactly those k positive coordinates.

Proof. Here $c_1 = 0$ and $c_k = -(k-1)a_k$ for $k \geq 2$. Since $k^{-1} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines in (28) have intercepts $c_k - H(k-1)(2k-1)/6$. The intersection of lines k and $k+1$ is therefore $t = \tau_k$. Extend c_k to the displayed smooth function $c(x)$ for $x \geq 1$. Direct differentiation gives

$$0 < c''(x) = \frac{x - 1/2}{x^2} \leq \frac{1}{2}.$$

Integrating this bound twice over unit intervals yields $c_{k+2} - 2c_{k+1} + c_k \leq 1/2$. Hence

$$\tau_{k+1} - \tau_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $\tau_{k-1} < t < \tau_k$, summing consecutive differences $\lambda_{j+1}(t) - \lambda_j(t) = t - \tau_j$ shows that line k strictly exceeds every earlier and later line. The fields are strictly ordered, so its unique best support is $\{1, \dots, k\}$. Apply Theorem 6.2 to obtain the finite- N conclusion. $\square$

Example 6.4 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$, the three competing maximal scores are

$$0, \quad t - a_2, \quad 2(t - a_3) - H/3.$$

If $H > 6(a_2 - a_3)$, the edge with support $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2.$$

For every fixed t in this interval, all global modes of the tilted law lie on that edge for sufficiently large N . Without the field, no proper face holds a mode in the critical window (Corollary 3.7).

7 Verification and open questions

The proofs above do not rely on computation. The Python scripts for this paper use only the standard library and check the results at finite size. `verify_geometry.py` compares 6,733 exact rational probabilities in 237 distributions with a dynamic program, and 158 of the distributions also with exhaustive enumeration. It checks normalization, means, covariances, and the values at $p = 0, 1$ for $d = 2, 3, 4$. `verify_simplex_threshold.py` checks the positive refresh formula, the balancing transfers, the finite-size edge mode, and the crossing expansions. `verify_frontier_boundary.py` and `verify_frontier_modes.py` check the uniform comparison, the rare-coordinate formulas, the spatial profiles, and the weak-field results against finite calculations. Finally, `verify_vertex_completion.py` makes 462 exact identity comparisons, 200 exact checks of the elasticity bounds, 16 root-bracket checks, and 4 large cases with growing rare counts. The output is stored under `data/`. The high-precision decimal checks are numerical and do not replace the proofs.

For a fixed support size, the leading near-vertex crossing is uniform over all of $A = o(b)$, but we do not have a uniform higher-order expansion in that regime. Other open questions are a growing number of colors, nonuniform switching, and the full mode diagram at finite N outside the critical window. Example 3.8 and Proposition 5.5 show that the asymptotic results cannot be strengthened in the obvious ways.

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